Data Science Project Scaffolding Guide

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Chapter 0: Overall Introduction

Objective

This guide is designed to help students complete data science projects independently by providing step-by-step scaffolding support. The focus is on guiding students through questions and hints rather than providing direct answers.

Core Principles

1. **Diagnose before guiding:** Identify where the student is stuck (conceptual misunderstanding, method selection, coding issues, result interpretation, or project planning) before providing hints.
2. **Do not provide final answers directly:** Encourage students to think critically and explore solutions.
3. **Use guiding questions and checkpoints:** Each project stage contains diagnostic questions, hints, and common mistakes to help students self-correct.
4. **Gradual hinting:** Start with conceptual hints, then checklists, and finally procedural or pseudocode hints if needed.

How to Use This Guide

- **Step 1:** Student asks a question.
- **Step 2:** The chatbot searches the uploaded PDF for relevant sections using keywords or semantic similarity.
- **Step 3:** The chatbot identifies the likely project stage and type of problem.
- **Step 4:** Based on the retrieved section, the chatbot first asks guiding questions, then provides hints, escalating only if the student remains stuck.
- **Step 5:** Repeat until the student can make progress independently.

Expected Outcomes

After following this scaffolding approach, students should be able to:

- Understand project objectives and plan effectively.
- Explore and preprocess data systematically.
- Select and implement multiple methods for comparison.

- Evaluate and interpret results accurately.
- Communicate findings clearly in reports or presentations.

Structure of the Guide

1. **Project Planning & Problem Formulation**
2. **Data Understanding & Exploration (EDA)**
3. **Data Preprocessing**
4. **Method Selection & Model Planning**
5. **Model Execution & Parameter Tuning**
6. **Model Evaluation & Comparison**
7. **Result Interpretation & Reporting**
8. **Project Examples**
9. **Keyword / Error Index**

Chapter 1: Project Planning & Problem Formulation

Objective

- Clearly define the project goal and scope.
- Identify the type of data science task (hypothesis testing, regression, classification, clustering, association rules, outlier detection, recommendation, natural language processing, computer vision, etc.).
- Establish measurable success criteria and evaluation metrics.
- Outline potential solution approaches for comparison.

Expected Outputs

- Project goal statement.
- Description of required data and features.
- Initial solution ideas or methods to explore. Must compare multiple solutions for solid comparison and validations.
- Chosen evaluation metrics or criteria for success. Figure out an evaluation plan.

Common Obstacles

- Unclear or vague project goal.
- Difficulty determining task type (e.g., regression vs. classification).
- Proposed solutions are not appropriate ones for the proposed problems or questions.
- Confusion over which evaluation metric best reflects the project objective.
- Lack of multiple solution options for comparison.

Guiding Questions

- What is the main question or problem your project aims to solve?
- Is your task for the purpose of predictions or pattern discovery
- Is your task predicting numerical values, classifying categories, clustering, or recommending items?
- What is your target variable or outcome of interest?

- Which evaluation metrics would reflect success for this task?
- What possible methods or algorithms could you try?
- How can you define a baseline solution to compare against?

Typical Mistakes

- Defining a goal that is too broad or vague (e.g., “analyze data” without specifying an objective).
- Proposed solutions are not appropriate ones for the proposed problems or questions.
- Choosing evaluation metrics that do not match the project task (e.g., using accuracy for imbalanced classification).
- Planning to implement only one solution method without comparison.
- Ignoring the need for a baseline model to measure improvement.

Cognitive Misconceptions

- Believing that a broad goal like “analyze data” is sufficient without specifying measurable outcomes.
- Confusing task type (e.g., thinking a classification problem is regression).
- Assuming the first solution idea is always correct or sufficient.
- Overlooking the need for baseline methods to evaluate improvement.
- Misunderstanding evaluation metrics, thinking higher numbers are always better without context.

Tips for Scaffolding Chatbot

- If a student asks “I don’t know where to start,” first clarify the project objective.
- Ask about task type and target variable before suggesting any method.
- Encourage the student to list at least 2-3 possible solution approaches.
- Use evaluation metrics as a checkpoint to make sure the project is measurable.

Socratic Question Prompts

1. Questions of Clarification

- “What exactly is the main goal of your project?”
- “Can you define your target variable clearly?”
- “Could you give an example of the type of output you expect?”

2. Questions that Probe Assumptions

- “Why do you assume this task type (regression/classification) fits your problem?”
- “What assumptions are you making about your data or variables?”
- “Are you assuming any relationships between features that may not exist?”

3. Questions that Probe Reasons and Evidence

- “What evidence supports choosing this project goal?”
- “How do you know this variable is relevant to your prediction?”
- “What data or prior knowledge guided your baseline selection?”

4. Questions about Viewpoints and Perspectives

- “What would someone who disagrees with your task formulation say?”
- “Are there alternative ways to define success for this project?”
- “How might a domain expert approach this differently?”

5. Questions that Probe Implications and Consequences

- “If you select this goal, what is the expected outcome or impact?”
- “What could happen if your project objectives are too vague?”
- “How will your evaluation approach influence interpretation of results?”

6. Questions about the Question (Meta-Questions)

- “Why is defining a clear project goal important?”
- “How could you verify that your problem formulation is meaningful?”
- “How does clarifying this question guide all subsequent steps?”

Chapter 2: Data Understanding & Exploration (EDA)

Objective

- Gain familiarity with the dataset structure, variables, and their types.
- Identify patterns, trends, and potential data issues (missing values, outliers, redundancy).
- Generate insights that will inform preprocessing and modeling decisions.

Expected Outputs

- Data summary tables (number of rows, columns, variable types, missing values).
- Correctly identify the data types of the variables
- Basic descriptive statistics for each variable.
- Visualizations of distributions, correlations, and relationships.
- Initial identification of potential data issues (e.g., outliers, inconsistencies).
- Initial removal of useless features

Common Obstacles

- Uncertainty about which variables are relevant or redundant.
- Some variables are direct indicators of the target variables to be predicted, but students are not sure whether they should be removed
- Incorrectly identify data types, especially when it comes to the variables with integer values.
- Use wrong approaches to measure correlation or dependency between variables.
- Difficulty identifying outliers or unusual data points.
- Confusion over appropriate visualizations for different data types.
- Overlooking missing values or misinterpreting patterns.

Guiding Questions

- What are the types and ranges of each variable?

- What are the data types of the variables? Are they numerical or nominal variables if you observe integers in the variable?
- Are there missing values? How significant are they?
- Are there any outliers or unusual observations?
- Do any variables appear redundant or highly correlated?
- Did you correctly figure out the data types of the variables before selecting the method to measure correlations or dependency between variables?
- Which plots or visualizations best illustrate the data distributions and relationships?
- Are there initial insights that suggest which features might be important for modeling?

Typical Mistakes

- Ignoring missing values or treating them inconsistently.
- Not checking for outliers or data entry errors.
- Failing to explore correlations and relationships between variables.
- Using inappropriate plots or summary statistics for certain data types.
- Considering the variable as a numerical one as long as there are integer values in the variable. However, it is possible to be encoded integers rather than numerical values.
- Selecting wrong methods to measure correlations or dependency between variables.
- Some variables are direct indicators for the target variables to be predicted, students failed to identify them and remove them.

Cognitive Misconceptions

- Assuming the dataset is “clean” and free from errors or outliers.
- Believing visualizations are optional and descriptive statistics are unnecessary.
- Misinterpreting correlation as causation.
- Ignoring the importance of exploring distributions before modeling.
- Thinking missing values or unusual entries can be safely ignored.

Tips for Scaffolding Chatbot

- If a student asks “I don’t understand the data,” first prompt them to describe the variables and types.
- Encourage students to confirm the data types of the variables before running correlation or dependency analysis.
- Encourage plotting distributions and correlations before any preprocessing.
- Guide students to check for missing values and outliers systematically.
- Ask which variables might be candidates for feature selection or reduction.
- Use insights from EDA to lead into preprocessing discussions in the next chapter.

Socratic Question Prompts

1. Questions of Clarification

- “What type of each variable are you dealing with?”
- “Can you explain what missing values or outliers you found?”
- “Could you describe the distribution of this feature?”

2. Questions that Probe Assumptions

- “Why do you assume this variable is informative?”
- “Are you assuming the data is clean and representative?”
- “What assumptions are made about relationships between features?”

3. Questions that Probe Reasons and Evidence

- “What evidence shows that this feature correlates with the target?”
- “How did you detect outliers or anomalies?”
- “Which statistical summaries support your understanding of the data?”

4. Questions about Viewpoints and Perspectives

- “How might someone else interpret the same data differently?”
- “Could different visualizations lead to alternative insights?”

- “What would a domain expert focus on first?”

5. Questions that Probe Implications and Consequences

- “If you ignore outliers, what could happen to your model?”
- “What is the consequence of not analyzing correlations?”
- “How would misinterpreting distributions affect preprocessing?”

6. Questions about the Question (Meta-Questions)

- “Why is exploring and understanding data crucial before modeling?”
- “How could you ensure your data insights are valid?”
- “Which questions should guide your EDA priorities?”

Chapter 3: Data Preprocessing

Objective

- Clean and prepare the data for modeling.
- Handle missing values, remove redundant features, and select relevant features.
- Split data properly for training, validation, and testing (hold-out or N-fold cross-validation).
- Apply transformations or scaling if necessary.

Expected Outputs

- Cleaned and preprocessed dataset ready for modeling.
- Documented preprocessing steps, including missing value treatment and feature selection/reduction.
- Train/test or cross-validation splits defined and reproducible.

Common Obstacles

- Uncertainty about which features to keep or drop.
- Difficulty deciding how to handle missing values.
- Difficulty deciding which methods to be used to measure data correlations
- Overlooking redundant features or correlated variables.
- Misunderstanding how to properly split data for training and validation.
- Difficulty selecting the data splits methods

Guiding Questions

- Which features are likely irrelevant or redundant?
- Are there missing values? How should they be handled (mean, median, mode, imputation)?
- Did you correctly figure out the data types of the variables before selecting the method to measure correlations or dependency between variables?
- Do any features need transformation or scaling?

- What is the best strategy for splitting the dataset (hold-out, K-fold cross-validation)?
- How can you ensure that data leakage does not occur between training and testing sets?

Typical Mistakes

- Including redundant or irrelevant features in the model.
- Inconsistent treatment of missing values across train/test datasets.
- Selecting wrong methods to measure correlations or dependency between variables.
- Failing to apply scaling when required (e.g., for distance-based algorithms).
- Selecting wrong data splits method, or deciding to use both of them.
- Mixing training and testing data or incorrectly performing cross-validation.
- Not documenting preprocessing steps, making reproducibility difficult.

Cognitive Misconceptions

- Believing that dropping features or rows without analysis will not affect model results.
- Assuming missing values can be filled arbitrarily without understanding their impact.
- Thinking scaling or normalization is only for some models and can be skipped.
- Confusing correlation with redundancy, dropping variables incorrectly.
- Assuming train/test splits do not need to reflect real-world data distribution.

Tips for Scaffolding Chatbot

- You are not responsible for coding or debugging codes. We only provide suggestions without coding assistance.
- If a student asks “What should I do with missing values?” first ask what percentage is missing and whether it’s random.
- If a student asks “how to measure correlations?” first ask the data types of the two variables.
- Encourage students to justify feature selection instead of dropping features blindly.

- Guide students to perform preprocessing **only on the training set** and then apply transformations to the test set.
- Ask about potential correlations and multicollinearity issues before dropping features.
- Remind students to keep track of preprocessing steps—they are part of the project documentation.

Socratic Question Prompts

1. Questions of Clarification

- "Can you explain why each preprocessing step is needed?"
- "What exactly do you mean by feature selection or scaling here?"
- "Could you illustrate how you plan to handle missing values?"

2. Questions that Probe Assumptions

- "Why do you assume dropping this feature won't harm your model?"
- "Are you assuming all missing values are random?"
- "What assumptions are embedded in your train/test split?"

3. Questions that Probe Reasons and Evidence

- "What evidence supports this imputation method?"
- "How do you know this feature is redundant?"
- "Why did you choose this scaling method for these variables?"

4. Questions about Viewpoints and Perspectives

- "How would a different preprocessing strategy affect your results?"
- "Could someone else argue that keeping these features is better?"
- "What are alternative ways to handle missing data?"

5. Questions that Probe Implications and Consequences

- "If you improperly split data, what could go wrong?"
- "What effect might removing correlated features have on model performance?"
- "How will scaling or normalization impact distance-based algorithms?"

6. Questions about the Question (Meta-Questions)

- “Why is preprocessing critical for downstream modeling?”
- “How could you check that your preprocessing choices are appropriate?”
- “Which preprocessing decisions matter most for reproducibility?”

Chapter 4: Method Selection & Model Planning

Objective

- Identify suitable algorithms or methods based on the project task (regression, classification, clustering, recommendation, NLP, computer vision, etc.).
- Plan multiple solution approaches to allow comparison.
- Consider baseline methods for benchmarking.
- Document reasoning behind method selection.
- Deal with special issues, such as imbalance issues in classification

Expected Outputs

- List of selected methods or algorithms (at least 2–3 for comparison).
- Justification for each method choice.
- Baseline method(s) identified for initial evaluation.
- Notes on potential strengths, weaknesses, and applicability.

Common Obstacles

- Difficulty choosing methods that match the task type.
- Ignore the imbalance issue
- Tendency to try only one method without comparison.
- Unclear reasoning behind method choice.
- Overlooking baseline or simpler approaches.

Guiding Questions

- What is the task type: regression, classification, clustering, recommendation, or other?
- Which methods are commonly used for this task?
- Are the labels balanced in your data if you run the classification tasks?
- What are the advantages and limitations of each method?
- Have you defined a simple baseline method for comparison?

- How will you compare the performance of different methods?
- Are there domain-specific considerations that affect method choice (e.g., NLP, image, or audio data)?

Examples of Methods by Task

- **Regression:** Linear regression, non-linear regression, SVR, regression trees, random forest, neural networks
- **Classification:** KNN, Naïve Bayes, decision trees, SVM, logistic regression, neural networks, assembling methods (bagging, random forest, adaBoosting, boosting, etc.)
- **Clustering:** K-means, hierarchical clustering, DBSCAN, density-based clustering.
- **Anomaly Detection:** Isolation forest, local outlier factor
- **Association Rules:** Apriori, FP-Growth
- **Advanced Applications:** NLP models (text classification, TF-IDF, RNN), recommender systems (collaborative filtering, matrix factorization, content-based approaches), computer vision (image/video recognition, face detection)

Typical Mistakes

- Selecting only one method and skipping comparison.
- Using methods incompatible with the data type or task.
- Ignoring baseline methods or simple models.
- Failing to document why each method was chosen.
- Not considering method scalability or complexity.

Cognitive Misconceptions

- Assuming the most complex or advanced algorithm is always the best.
- Believing one method is enough without comparison.
- Thinking domain-specific methods are optional even if they affect results.
- Misunderstanding baseline methods as optional or trivial.
- Confusing model accuracy with real-world usefulness or significance.

Tips for Scaffolding Chatbot

- If a student asks “Which method should I use?” first ask about the task type and data characteristics.
- If a student asks “Which solutions should be used for imbalance?” first ask about the categories of solutions for imbalance issues and which one they want to adopt.
- Encourage listing multiple candidate methods before implementation.
- Ask the student to explain their reasoning for choosing each method.
- Remind the student to define at least one baseline for comparison.
- Guide students to think about evaluation metrics when planning methods.

Socratic Question Prompts

1. Questions of Clarification

- “Can you clarify why this method is suitable for your task?”
- “What do you mean by comparing multiple solutions?”
- “Could you specify the baseline method you plan to use?”

2. Questions that Probe Assumptions

- “Why do you assume this algorithm will perform best?”
- “Are you assuming your data meets the method’s requirements?”
- “What assumptions are you making about feature relevance?”

3. Questions that Probe Reasons and Evidence

- “What evidence suggests this algorithm is effective?”
- “Why do you think these methods are worth comparing?”
- “How did previous studies or domain knowledge guide your choices?”

4. Questions about Viewpoints and Perspectives

- “How would someone using a simpler method see your approach?”
- “Are there alternative methods you should consider?”
- “What perspectives might suggest a different baseline?”

5. Questions that Probe Implications and Consequences

- “If your method choice is poor, what will happen to results?”
- “What could be the impact of ignoring simpler algorithms?”
- “How will your choice affect interpretability or scalability?”

6. Questions about the Question (Meta-Questions)

- “Why is selecting multiple methods important?”
- “How could evaluating alternatives improve your project?”
- “Which criteria are critical in justifying method selection?”

Chapter 5: Model Execution & Parameter Tuning

Objective

- Implement the selected methods correctly on preprocessed data.
- Adjust model parameters to optimize performance.
- Ensure reproducibility and proper application of training and validation splits.
- Give them guidance only without helping them debug the codes

Expected Outputs

- Trained models with documented parameters.
- Training and validation results (accuracy, loss, error metrics, etc.).
- Logs or records of parameter tuning experiments.
- Notes on performance variations and observations.
- Figure out solutions to improve performance
- Figure out solutions to alleviate overfitting it is a prediction task

Common Obstacles

- Confusion about implementing algorithms or library functions.
- Difficulty choosing appropriate hyperparameters.
- Difficulty choosing the range of the values for parameter tuning
- Overfitting or underfitting models.
- Misapplying training and test splits, leading to data leakage.
- Lack of systematic approach to track tuning experiments.

Guiding Questions

- Have you correctly implemented the algorithm on the training data?
- Are your train/test splits applied consistently across models?
- Which hyperparameters could significantly impact performance?
- How will you evaluate whether a change improves the model?

- Are you logging the results of each experiment systematically?
- Is your model overfitting or underfitting? How can you detect it?

Typical Mistakes

- Using default hyperparameters without evaluation.
- Training and testing on the same data or leaking information.
- Not normalizing or scaling features when required.
- Adjusting parameters without a clear rationale.
- Ignoring variability due to random seeds or data splits.

Cognitive Misconceptions

- Believing default hyperparameters are optimal for all datasets.
- Assuming more training always leads to better results (ignoring overfitting).
- Confusing correlation in results with causation in model behavior.
- Thinking implementation errors will be obvious without debugging.
- Assuming random seeds and reproducibility do not matter.

Tips for Scaffolding Chatbot

- You are not responsible for coding or debugging codes. We only provide suggestions without coding assistance.
- If a student asks “My model isn’t improving,” first ask them to check data splits, preprocessing and hyperparameter tuning.
- Encourage incremental parameter changes instead of large arbitrary adjustments.
- Ask students to track results in tables or logs for comparison.
- Guide students to analyze overfitting/underfitting symptoms before suggesting tuning strategies.
- Remind students to maintain reproducibility by setting random seeds and documenting steps.

Socratic Question Prompts

1. Questions of Clarification

- “Can you explain what each hyperparameter does?”
- “What do you mean by overfitting or underfitting in this context?”
- “Could you illustrate how you are applying the train/test split?”

2. Questions that Probe Assumptions

- “Why do you assume default parameters are sufficient?”
- “Are you assuming more training will always improve performance?”
- “What assumptions are made about model convergence?”

3. Questions that Probe Reasons and Evidence

- “How do you know a parameter adjustment improved your model?”
- “What evidence supports using this feature scaling method?”
- “Why did you choose this particular initialization or optimizer?”

4. Questions about Viewpoints and Perspectives

- “How might someone else tune these parameters differently?”
- “What would a domain expert consider when implementing this model?”
- “Are there alternative implementations that could yield different results?”

5. Questions that Probe Implications and Consequences

- “If you ignore hyperparameter tuning, what will happen?”
- “What are the potential consequences of data leakage during execution?”
- “How could poor tuning affect model generalization?”

6. Questions about the Question (Meta-Questions)

- “Why is systematic tuning important?”
- “How could you evaluate whether your execution process is sound?”
- “Which steps ensure your experiments are reproducible?”

Chapter 6: Model Evaluation & Comparison

Objective

- Assess the performance of multiple models or methods.
- Compare different approaches using appropriate metrics.
- Identify strengths, weaknesses, and trade-offs between models.

Expected Outputs

- Evaluation tables or charts comparing model performance.
- Interpretation of metrics (accuracy, precision, recall, F1-score, RMSE, AUC, etc.).
- Notes on model selection rationale.
- Recommendations for next steps or improvements.

Common Obstacles

- Misunderstanding metrics and their implications.
- Focusing on only one metric without considering others.
- Difficulty comparing models with different strengths or weaknesses.
- Overlooking baseline comparisons.
- Ignoring cross-validation results or variability.
- Not satisfied with models with accuracy lower than 90%

Guiding Questions

- Which metrics are most relevant for your task?
- How does each model perform on different metrics?
- Are there trade-offs between models (e.g., precision vs. recall)?
- Does any model outperform the baseline?
- Are the results consistent across cross-validation folds?
- What do the evaluation results suggest about feature selection, parameter tuning, or method choice?

Typical Mistakes

- Using an inappropriate metric for the task (e.g., accuracy on imbalanced classification).
- Comparing models without considering variability or confidence intervals.
- Ignoring simpler baseline models when evaluating performance.
- Overfitting to the validation set during tuning.
- Misinterpreting results due to lack of domain context.

Cognitive Misconceptions

- Believing a single metric is sufficient to judge model performance.
- Thinking that higher accuracy automatically means a better model in imbalanced tasks.
- Confusing training/validation performance with real-world generalization.
- Ignoring baseline performance and overestimating model gains.
- Assuming cross-validation folds always give identical results.

Tips for Scaffolding Chatbot

- If a student asks "Which model is best?" first prompt them to check metrics relevant to the task.
- Encourage comparing multiple metrics rather than a single number.
- Ask about consistency across cross-validation or multiple runs.
- Guide students to consider baseline performance before concluding.
- Remind students to connect evaluation results back to project objectives.

Socratic Question Prompts

1. Questions of Clarification

- "Can you define the metric you are using clearly?"
- "What exactly does this performance number represent?"
- "Could you explain the difference between training and validation results?"

2. Questions that Probe Assumptions

- “Why do you assume one metric is sufficient?”
- “Are you assuming model performance will generalize to new data?”
- “What assumptions are made in cross-validation?”

3. Questions that Probe Reasons and Evidence

- “What evidence shows that this model is better than the baseline?”
- “How do you justify preferring one model over another?”
- “Why do you trust the evaluation metric for this task?”

4. Questions about Viewpoints and Perspectives

- “How would someone using another metric interpret your results?”
- “What would a critic say about your comparison?”
- “Could a different baseline change your conclusion?”

5. Questions that Probe Implications and Consequences

- “If you misinterpret evaluation metrics, what could happen?”
- “What are the consequences of ignoring cross-validation variance?”
- “How will your choice of model affect deployment or usability?”

6. Questions about the Question (Meta-Questions)

- “Why is comparing multiple models important?”
- “How could you ensure your evaluation is meaningful?”
- “Which metrics are most relevant to your project objectives?”

Chapter 7: Result Interpretation & Reporting

Objective

- Summarize the project results in a clear, logical manner.
- Interpret the outcomes of different models and methods.
- Communicate insights and conclusions effectively in written reports or presentations.
- Identify limitations and propose improvements or next steps.

Expected Outputs

- Written report or presentation with structured sections: Introduction, Methods, Results, Discussion, Conclusion.
- Clear explanation of model performance and comparisons.
- Insights about key features, patterns, or trends discovered.
- Recommendations for future work or project extensions.

Common Obstacles

- Difficulty explaining results in plain language.
- Confusing metric values with practical significance.
- Ignoring differences between models or misinterpreting comparisons.
- Failing to connect findings to the original project objectives.
- Overlooking limitations, assumptions, or potential biases.

Guiding Questions

- What are the main findings from your analysis?
- How do the results of different methods compare?
- Which features or factors contributed most to the results?
- Are there surprising or unexpected results? How can you explain them?
- How do your results address the project objectives?

- What are the limitations of your approach, and how could future work improve on them?

Typical Mistakes

- Presenting numbers or charts without interpretation.
- Ignoring the context of the data or project goal.
- Overstating conclusions or ignoring limitations.
- Failing to justify why one method is preferred over another.
- Neglecting to propose future improvements or experiments.

Cognitive Misconceptions

- Thinking reporting numbers alone is equivalent to analysis.
- Believing all observed patterns are meaningful without considering noise or bias.
- Overgeneralizing conclusions beyond the dataset scope.
- Assuming limitations or assumptions are unimportant for interpretation.
- Confusing statistical significance with practical significance.

Tips for Scaffolding Chatbot

- If a student asks “What do these results mean?” start by asking them to summarize the key metrics.
- Encourage linking results back to the project question or objective.
- Guide students to compare multiple models before drawing conclusions.
- Ask about limitations and potential biases to promote critical thinking.
- Prompt students to highlight actionable insights rather than just reporting numbers.

Socratic Question Prompts

1. Questions of Clarification

- “What do you mean by this result or pattern?”
- “Can you explain your conclusion in simple terms?”
- “Could you illustrate how this insight answers your project question?”

2. Questions that Probe Assumptions

- “Why do you assume this pattern is meaningful?”
- “Are you assuming causation from correlation?”
- “What assumptions underlie your interpretation?”

3. Questions that Probe Reasons and Evidence

- “What evidence supports your conclusions?”
- “How do your evaluation metrics justify your interpretation?”
- “Why do you trust these results over alternative explanations?”

4. Questions about Viewpoints and Perspectives

- “How might someone else interpret these findings?”
- “What alternative explanations exist for your observations?”
- “Are there other stakeholders who would see this differently?”

5. Questions that Probe Implications and Consequences

- “What impact would your findings have in practice?”
- “What are the consequences if these results are misinterpreted?”
- “How could these insights guide future projects?”

6. Questions about the Question (Meta-Questions)

- “Why is interpreting results carefully important?”
- “How could you ensure your report is clear and actionable?”
- “Which conclusions are most critical to communicate?”

Chapter 8: Project Examples

Objective

- Illustrate the application of the previous chapters in real project scenarios.
- Demonstrate step-by-step scaffolding for common types of data science projects.
- Show how multiple methods can be compared and evaluated.

Example 1: House Price Prediction (Regression)

Project Steps:

1. **Planning:** Predict house prices based on features like area, number of rooms, and location.
2. **Data Understanding & EDA:** Check distributions of house prices, detect outliers, examine correlations.
3. **Preprocessing:** Handle missing values, remove redundant features, normalize numeric variables, split data into train/test sets.
4. **Method Selection:** Linear regression, regression trees, SVR; include baseline mean predictor.
5. **Model Execution & Tuning:** Train models, adjust hyperparameters (e.g., tree depth, SVR kernel parameters).
6. **Evaluation & Comparison:** RMSE, MAE, and R^2 metrics; compare models and baseline.
7. **Interpretation & Reporting:** Identify key factors influencing price, explain model differences, propose improvements.

Scaffolding Focus:

- Ask students to identify outliers before model training.
- Prompt them to justify method choices and hyperparameters.
- Encourage comparison with baseline.

Example 2: Sentiment Analysis (NLP Classification)

Project Steps:

1. **Planning:** Classify text reviews as positive or negative.
2. **Data Understanding & EDA:** Analyze review lengths, word frequency distributions, class balance.
3. **Preprocessing:** Tokenization, stopword removal, vectorization (TF-IDF), train/test split.
4. **Method Selection:** Logistic regression, Naive Bayes, simple neural network; include baseline (most frequent class).
5. **Model Execution & Tuning:** Train models, adjust regularization or network layers, monitor loss curves.
6. **Evaluation & Comparison:** Accuracy, precision, recall, F1-score; compare models and baseline.
7. **Interpretation & Reporting:** Highlight important words or phrases, explain model performance, suggest improvements.

Scaffolding Focus:

- Guide students to balance classes or consider metrics beyond accuracy.
- Prompt justification of preprocessing choices.
- Encourage interpretation of feature importance.

Example 3: Recommender System (Collaborative Filtering)

Project Steps:

1. **Planning:** Recommend movies to users based on past ratings.
2. **Data Understanding & EDA:** Check sparsity of rating matrix, distribution of ratings, user/item activity.
3. **Preprocessing:** Handle missing ratings, normalize, split train/test sets.
4. **Method Selection:** User-based CF, item-based CF, matrix factorization; include baseline (average rating).

5. **Model Execution & Tuning:** Train models, adjust neighborhood size or latent factors.
6. **Evaluation & Comparison:** RMSE, MAE, top-N recommendation precision/recall; compare models to baseline.
7. **Interpretation & Reporting:** Identify strengths of each method, discuss sparsity effects, propose enhancements.

Scaffolding Focus:

- Ask students to justify method selection.
- Guide evaluation using multiple metrics.
- Encourage discussion of trade-offs between methods.

Tips for Scaffolding Chatbot

- When a student asks about a project, first identify which example aligns with their task.
- Use guiding questions from previous chapters to prompt critical thinking.
- Encourage students to explore multiple methods, justify choices, and document results.
- Highlight common pitfalls seen in similar projects (e.g., overfitting, ignoring baseline, misunderstanding metrics).

Chapter 9: Keyword / Error Index

Objective

- Provide a structured index of keywords, concepts, and common student errors.
- Facilitate fast retrieval of relevant sections in the PDF using semantic search or keyword matching.
- Support the chatbot in identifying the type of problem (conceptual, method, code, interpretation, planning).

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1. Project Planning & Problem Formulation

Keywords: project goal, task type, objective, target variable, evaluation metric, baseline, solution approach
Common Student Errors: vague goals, one solution only, irrelevant metrics, missing baseline

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2. Data Understanding & EDA

Keywords: variable type, descriptive statistics, missing values, outliers, correlation, distribution plots, trends, imbalance
Common Student Errors: ignoring missing values, failing to explore correlations, using wrong plots, misinterpreting data

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3. Data Preprocessing

Keywords: missing value imputation, feature selection, feature reduction, scaling, normalization, train/test split, cross-validation, data leakage, imbalance
Common Student Errors: inconsistent missing value handling, retaining redundant features, improper train/test split, ignoring preprocessing logs

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4. Method Selection & Model Planning

Keywords: algorithm choice, regression, classification, clustering, anomaly detection, recommendation, baseline method, method comparison, imbalance

Common Student Errors: incompatible methods, only one method, ignoring baseline, unclear justification

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5. Model Execution & Parameter Tuning

Keywords: model training, hyperparameter tuning, overfitting, underfitting, reproducibility, train/test split, experiment logging
Common Student Errors: default parameters without evaluation, data leakage, lack of systematic parameter tracking, scaling issues

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6. Model Evaluation & Comparison

Keywords: metrics, accuracy, precision, recall, F1-score, RMSE, AUC, cross-validation, performance comparison, trade-offs
Common Student Errors: inappropriate metrics, ignoring baseline, single-metric focus, misinterpreting results, overfitting validation

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7. Result Interpretation & Reporting

Keywords: summary, explanation, insights, findings, limitations, assumptions, recommendations, report structure, visualizations
Common Student Errors: reporting numbers without interpretation, overstating conclusions, ignoring limitations, poor connection to project objectives

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8. Project Examples

Keywords: regression, classification, NLP, recommender system, collaborative filtering, matrix factorization, linear regression, decision tree, SVR, logistic regression, Naive Bayes
Common Student Errors: skipping preprocessing steps, ignoring baseline, misunderstanding evaluation metrics, overfitting

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Tips for RAG Chatbot

- When a student asks a question, first match keywords or semantic embeddings to identify the likely chapter and section.

- Check if the issue is conceptual, method-related, coding, result interpretation, or planning.
- Use guiding questions, hints, and common errors as scaffolding prompts.
- Escalate gradually: start with conceptual hints, then procedural hints, then pseudocode if needed.

Chapter 10: Trigger Matrix – Usage Guidance

Purpose:
The Trigger Matrix in this chapter is designed to serve as a **diagnostic routing guide** for data science projects. It maps common student questions, keywords, and errors to the corresponding project phase and chapter, enabling focused guidance and scaffolding.

How to Use:

1. **Identify the Phase:** Look at the student's question or problem description.
2. **Match Keywords or Common Errors:** Scan the "Keywords" and "Common Errors" columns to find the best fit.
3. **Determine Diagnosis Type:** Each row has a "Diagnosis Type" (Concept / Method / Planning / Evaluation / Code / Interpretation), which indicates the kind of guidance needed.
4. **Route to Chapter:** Use the "Route to Chapter" column to locate the corresponding chapter that contains Socratic prompts and cognitive misconceptions relevant to the student's situation.
5. **Use Example Questions as Reference:** The "Example Student Question" column can help you interpret how typical student queries are linked to the matrix.

Tips:

- Multiple keywords or errors may apply; choose the row that best matches the student's difficulty.
- This matrix is meant to **guide thought, not provide direct answers**—use the corresponding chapter's Socratic prompts to scaffold learning.
- You can expand the "Example Student Question" or "Keywords" columns over time to cover more student scenarios.
- Keep this chapter updated as new project types, common errors, or methods emerge.

Phase / Chapter	Diagnosis Type	Keywords	Common Errors	Example Student Question	Route to Chapter
Project Planning & Problem Formulation	Planning	project goal, target variable, problem definition, success criteria	vague objectives, unclear target, ignoring context	“What exactly should my project accomplish?”	1
Data Understanding & Exploration	Concept	descriptive statistics, distribution, correlation, outliers, missing values	misinterpreting distributions, ignoring outliers, misunderstanding correlation	“How do I know which features matter?”	2
Data Preprocessing	Method	feature selection, missing value handling, scaling, normalization, dimensionality reduction	dropping useful features, incorrect imputation, ignoring correlations	“How should I handle missing values?”	3
Method Selection & Model Planning	Method	algorithm choice, baseline method, model comparison, task type	choosing incompatible algorithms, ignoring baseline, unclear justification	“Which model should I use?”	4
Model Execution & Parameter Tuning	Method	hyperparameter tuning, train/test split, cross-validation, optimizer, overfitting	using default parameters blindly, ignoring overfitting, data leakage	“How do I tune the model properly?”	5

Phase / Chapter	Diagnosis Type	Keywords	Common Errors	Example Student Question	Route to Chapter
Model Evaluation & Comparison	Evaluation	metric selection, cross-validation, baseline comparison, performance analysis	overfitting metrics, misinterpreting validation, ignoring variance	“Which metric should I use to compare models?”	6
Result Interpretation & Reporting	Interpretation	insights, conclusions, causation vs correlation, visualization, reporting	misinterpreting correlation, overgeneralization, unclear communication	“How should I interpret these results?”	7